EVQ-Cosh: Closed-Form Frequency Allocation for Long-Context Extrapolation

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Abstract

Rotary position embedding is often extended at inference time, but its training-time geometric frequency allocation is usually left untouched. We formulate RoPE frequency allocation as a variational inverse problem, derive a broadband surrogate whose stationary density satisfies a second-order ODE, and obtain a closed-form family, EVQ-cosh, in which geometric RoPE is the degenerate $\tau = 0$ limit. Empirically, EVQ beats learnable adaptive positional encoding in DAPE-style extreme extrapolation ($128 \to 8$K: 333.7 vs. 455.3 PPL), showing that positional structure alone can determine extreme long-range behavior. More importantly, EVQ unlocks inference-time scaling: across six seeds, EVQ+YaRN reaches 100% retrieval at 8K, while Geo+YaRN remains at 61% to 65%; in a fair 10% mix comparison, EVQ+YaRN lowers PPL from 82.9 to 70.9 at 8K and from 157.7 to 107.5 at 16K. These results identify training-time frequency allocation as a missing foundation for long-context extrapolation.

1 Introduction

Long-context language modeling is often framed as an inference-time scaling problem: if rotary position embedding (RoPE) is compressed, interpolated, or rescaled carefully enough, a model trained at length L_{train} should extrapolate to substantially longer contexts [Chen et al., 2023, Peng et al., 2024, Ding et al., 2024]. This view has produced strong engineering baselines, but it leaves one design choice essentially untouched: the training-time geometric allocation of RoPE frequencies. Standard RoPE treats that allocation as fixed, even though long-range failure is ultimately mediated through frequency collisions in the learned representation.

We take the opposite starting point. Instead of asking only how to rescale positions at inference time, we ask how frequencies should be allocated during training. We formulate RoPE frequency allocation as a variational inverse problem, derive a broadband surrogate in which the stationary density satisfies a second-order ODE, and obtain a closed-form family, EVQ-cosh. In this family, geometric RoPE appears as the degenerate $\tau = 0$ limit rather than a generic optimum.

This change in viewpoint leads to two central empirical findings. First, in the PE-dominant extreme-extrapolation regime, EVQ beats learnable adaptive positional encoding with zero additional parameters: on the DAPE-style $128 \to 8$K setup, EVQ reaches 333.7 PPL while DAPE reaches 455.3. This is the cleanest setting in which positional structure itself is the dominant variable. Second, and more importantly, EVQ acts as the missing training-time foundation for inference-time scaling. Across six seeds, EVQ+YaRN reaches 100% retrieval at 8K, while Geo+YaRN remains at 61% to 65%; in a fair 10% mix comparison, EVQ+YaRN lowers PPL from 82.9 to 70.9 at 8K and from 157.7 to 107.5 at 16K.

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The paper's claim is therefore sharper than "EVQ improves RoPE." Our claim is that standard RoPE underperforms because its geometric frequency allocation is the wrong training-time foundation for long-context extrapolation. Inference-time scaling can only recover what the checkpoint geometry makes available. Once that geometry is improved, the strongest existing inference-time methods become significantly more effective.

This distinction matters operationally. The dominant deployment pattern in long-context LLMs is to keep the training checkpoint fixed and layer stronger inference-time scaling on top. If the checkpoint has already collapsed too much low-frequency resolution into a geometric schedule, then inference-time scaling is trying to stretch positions on top of the wrong substrate. Our thesis is that training-time frequency design is therefore a missing systems primitive, not merely a cleaner mathematical interpretation of RoPE.

Our contributions are:

- We derive a closed-form variational solution for RoPE frequency allocation and show that geometric RoPE is the $\tau = 0$ degenerate case of a broader optimal family.
- We show that closed-form EVQ beats learnable PE in DAPE-style extreme extrapolation, supporting the claim that PE quality itself can determine long-range behavior.
- We demonstrate that the strongest systems result is not merely EVQ over geometric RoPE, but EVQ unlocking YaRN: training-time and inference-time positional optimization are orthogonal and strongly complementary.
- We present a conservative empirical package in which the main claims rely on multi-seed text evidence, while larger-scale single-seed runs and cross-modal transfer results are treated as consistency checks rather than primary anchors.

2 Related Work

Rotary and relative positional methods. Transformers rely on positional structure to recover order information [Vaswani et al., 2017]. Popular alternatives include relative attention biases [Shaw et al., 2018, Raffel et al., 2020], ALiBi [Press et al., 2022], and RoPE / RoFormer [Su et al., 2024]. Our work is closest in object to RoPE, but unlike prior usage it treats frequency allocation itself as the quantity to optimize.

Inference-time context extension. Recent long-context methods often preserve the training checkpoint and modify positional behavior only at inference time, for example via positional interpolation, YaRN, LongRoPE, and related extension heuristics [Chen et al., 2023, Peng et al., 2024, Ding et al., 2024, Anil et al., 2023, Jin et al., 2024]. Survey work already frames positional design as a central obstacle in length extrapolation [Wang et al., 2024]. These methods are practically important and form the relevant systems baseline, but they do not directly optimize the training-time frequency geometry on which extrapolation ultimately relies. Our thesis is not that such methods are unimportant, but that they are operating on a checkpoint whose positional substrate may already be suboptimal.

Learnable and adaptive positional encoding. Another line of work introduces learnable or adaptive positional mechanisms, including Kerple, FIRE, and DAPE [Chi et al., 2023, Li et al., 2024, Zheng et al., 2024]. These methods are the closest comparison point for our PE-dominant experiments because they ask whether adaptive positional structure can improve extrapolation. Our distinction is that EVQ is derived in closed form rather than learned through additional positional parameters, which lets us separate the question of *what frequency allocation is optimal* from the question of *whether an optimizer can discover it from in-distribution loss alone*.

Evaluation for long-context behavior. Long-context evaluation is often split between synthetic retrieval-style diagnostics and broader long-context task suites such as LongBench and RULER [Bai et al., 2024, Hsieh et al., 2024]. Prior work has also shown that long contexts can fail even when relevant information is technically present [Liu et al., 2024]. We therefore use both PE-dominant extrapolation settings and retrieval-heavy long-context evaluation to separate mechanistic evidence from broader empirical support, and we treat incomplete larger-scale task evidence conservatively.

3 Theory and Method

3.1 RoPE as frequency allocation

Let $\phi \in [0, 1]$ parameterize log-frequency and define $\omega(\phi) = b^{-\phi}$. A discrete RoPE design is therefore equivalent to choosing a density $\rho(\phi)$ and then discretizing it by inverse CDF. Standard geometric RoPE corresponds to a uniform density over ϕ; EVQ instead derives ρ from a variational objective. This reparameterization moves the design space from a hand-written index rule to a continuous allocation problem.

3.2 Exact kernel and broadband surrogate

Under a distance prior $D(\Delta)$, the phase-collision kernel is

$$K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta)\, d\Delta. \tag{1}$$

Our only approximation step is a broadband surrogate,

$$K_{\text{app}}(\phi_1, \phi_2) \approx \alpha\delta(\phi_1 - \phi_2) + \beta \min(\phi_1, \phi_2), \tag{2}$$

which isolates a diagonal ridge term and a smooth off-diagonal covariance term. The δ term captures local ridge structure, while $\min(\phi_1, \phi_2)$ is the Green kernel of a one-dimensional second-order operator. After this step, the remaining derivation is exact.

3.3 Variational objective and exact ODE

The broadband functional takes the form

$$\mathcal{J}[\rho] = \frac{\alpha}{2} \int \rho(\phi)^2\, d\phi + \frac{\beta}{2} \iint \rho(\phi)\rho(\psi) \min(\phi, \psi)\, d\phi\, d\psi - \mu \int \rho(\phi) b^{-2\phi}\, d\phi. \tag{3}$$

Theorem 1. *After fixing the surrogate K_{app}, the stationary density satisfies*

$$\rho''(\phi) - \tau^2 \rho(\phi) = \gamma b^{-2\phi}, \qquad \tau = \sqrt{\beta/\alpha}, \tag{4}$$

with a closed-form solution consisting of a hyperbolic tether and a Fisher-driven forcing term.

This is the exact theory backbone used in the paper body: the surrogate is the only approximation, while the ODE and its closed-form family are exact conditional on that surrogate. The homogeneous branch supplies a global tether that redistributes density, while the forcing term preserves local Fisher utility. The practical significance of this decomposition is that it separates two qualitatively different requirements: a global allocation that decides where frequency mass should live, and a local utility term that prevents the solution from collapsing into a purely asymptotic long-range design. EVQ is therefore not an unconstrained long-range distortion of RoPE; it is the stationary compromise induced by the surrogate objective.

3.4 Closed-form EVQ and the geometric limit

The pure-tether branch yields an inverse-CDF warp

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \qquad u_k = \frac{k}{K}, \tag{5}$$

and inverse frequencies $b^{-\phi_k(\tau)}$.

Theorem 2. *EVQ-cosh converges smoothly to geometric RoPE as $\tau \to 0$. Therefore, geometric RoPE is the $\tau = 0$ degenerate limit of the EVQ family rather than a generic optimum.*

The discrete implementation then follows by evaluating $\phi_k(\tau)$ at evenly spaced quantiles u_k. In this form, EVQ changes no architecture component and requires no auxiliary positional network: it changes only the inverse-frequency initialization. This is the key methodological distinction from learnable PE approaches. The theory commits to a specific allocation map before optimization begins, rather than hoping that in-distribution training dynamics will discover the same map through extra positional parameters.

3.5 Practical EVQ instantiation

For a head dimension d_{head} and training length L, the implementation used throughout the paper is

$$\tau = \frac{d_{\text{head}}}{\sqrt{L}}, \qquad u_k = \frac{k}{K}, \qquad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}, \qquad K = \frac{d_{\text{head}}}{2}. \tag{6}$$

This construction is deliberately minimal: it modifies the inverse-frequency schedule but leaves the transformer, attention rule, optimizer, and inference stack unchanged. It is therefore a direct replacement for the standard geometric initialization rather than a learnable add-on.

In practice, the method reduces to four deterministic steps: choose L and d_{head}, set $\tau = d_{\text{head}}/\sqrt{L}$, evaluate the closed-form warp on uniform quantiles u_k, and replace the geometric inverse frequencies by $b^{-\phi_k(\tau)}$. There is no auxiliary positional loss, no extra MLP, and no optimizer dependence beyond the ordinary language-model training loop.

The quantile form also explains two implementation-level facts used later in the paper. First, $u_0 = 0$ implies $\phi_0(\tau) = 0$, so the first channel remains anchored at $\omega_0 = 1$, matching geometric RoPE exactly. Second, because the warp is small near the high-frequency end and grows toward the low-frequency end, EVQ does not require a separate "keep-r-channels-geometric" design choice in order to preserve useful short-range structure.

Practical recipe: EVQ-cosh initialization. Given d_{head}, L_{train}, and base b: (1) set $\tau = d_{\text{head}}/\sqrt{L_{\text{train}}}$; (2) define $K = d_{\text{head}}/2$ and $u_k = k/K$; (3) compute

$$\phi_k(\tau) = 1 - \frac{1}{\tau}\,\text{arcsinh}((1-u_k)\sinh\tau)\,;$$

(4) initialize inverse frequencies as $b^{-\phi_k(\tau)}$. This is the complete method used in the body experiments. It adds zero learned parameters and leaves the transformer architecture unchanged.

3.6 Waterbed and the scaling-law statement

Proposition 1. *Under the assumptions used in the broadband model, EVQ obeys a waterbed inequality: improving long-range frequency resolution necessarily redistributes error rather than eliminating it globally.*

This proposition explains why small in-range costs can coexist with large out-of-range gains. Geometric RoPE distributes density uniformly in log-frequency, while EVQ intentionally pays bounded high-frequency cost to relieve the low-frequency collision bottleneck.

We treat the length-dependent optimal temperature

$$\tau^*(L) = \frac{d_{\text{head}}}{\sqrt{L}} \tag{7}$$

as an empirical law / conjecture supported by multi-regime experiments, not as an unconditional theorem. This distinction keeps the theory honest: the exact claim is the surrogate-to-ODE chain, while the scaling law is a predictive summary of how the surrogate coefficients behave across training lengths.

Two consequences follow immediately. When L is short, $\tau^*(L)$ is large and the optimal schedule should depart visibly from the geometric allocation; this is precisely the PE-dominant regime in which DAPE-style comparisons are most revealing. When L grows, $\tau^*(L)$ shrinks and EVQ should converge smoothly back toward geometric RoPE, predicting bounded in-range cost together with a progressively narrower but still useful long-range correction. This is exactly the pattern that later appears in the multi-scale and continued-pretraining results.

4 Predictions and Mechanism

The theory predicts an asymmetric tradeoff. EVQ does not "push frequencies to the low end"; instead it redistributes channel density, expanding low-frequency spacing while compressing high-frequency

spacing. The anticipated gain is therefore concentrated in the regime where long-range discrimination is bottlenecked by low-frequency collisions. In the small-τ regime, the warp admits a smooth Taylor expansion,

$$\phi_k(\tau) \approx u_k - \frac{u_k(1-u_k)(2-u_k)}{6}\tau^2, \tag{8}$$

which shows that the deviation from geometric allocation grows predictably as the context shortens and $\tau^*(L)$ rises.

This leads to a waterbed interpretation of the empirical curves. Short-range costs should remain bounded because high-frequency channels are redundant for nearby positions, while long-range gains can be large because the low-frequency bottleneck is the dominant failure mode. This asymmetry explains why small in-range costs can coexist with large out-of-range improvements.

The same picture also predicts a base dependence. If frequencies with wavelength larger than L_{train} define a collision block of relative size $c \approx \ln L / \ln b$, then the amount of density available for useful reallocation scales with the complementary mass $1 - c$. In other words, EVQ should help most when the geometric schedule leaves a substantial low-frequency collision block to repair, and it should help much less when that block is already tiny.

The same mechanism predicts complementarity with inference-time scaling. YaRN rescales positions at inference time, which effectively compresses usable frequency spacing. If the training-time allocation is already geometric, this compression acts on an already crowded low-frequency regime. EVQ changes that geometry before inference-time scaling is applied, so EVQ expands low-frequency spacing precisely where YaRN would otherwise suffer the most. This is why the strongest systems result in the paper is not EVQ in isolation but EVQ combined with YaRN: inference-time scaling becomes substantially more effective once the checkpoint starts from a better training-time frequency substrate.

This mechanism also predicts how metric sensitivity should shift as models become stronger. Once coarse retrieval saturates, the remaining effect of better low-frequency geometry should appear more clearly in stricter precision-oriented metrics than in hit-rate alone. This is exactly the pattern observed in the larger-scale continued-pretraining result, where retrieval saturates at 100% but autoregressive exact recovery still separates EVQ sharply from the geometric baseline.

A final prediction concerns negative results. If the effective collision block is already tiny, there is little density to reallocate. This is why the low-base dead-zone experiments are theoretically useful rather than embarrassing: EVQ should *not* create gains when the collision-block analysis predicts little room for improvement. A negative result in that regime is therefore evidence that the mechanism is specific rather than overfit.

5 Experiments

5.1 Setup

All core experiments are from-scratch and keep the model architecture fixed. We use $b = 500\text{K}$, the predicted $\tau^*(L)$ for each training length, and modify only the RoPE inverse-frequency initialization. The primary datasets are TinyStories, FineWeb-Edu, and controlled passkey-mix settings; the main long-context evaluations are retrieval-style extrapolation and broader suites such as LongBench and RULER when available [Bai et al., 2024, Hsieh et al., 2024]. Architecturally, the models follow standard GPT-style decoder stacks [Radford et al., 2019], and no extra positional network is trained.

Primary body claims are anchored in multi-seed evidence. The core raw-PPL and passkey-mix comparisons use three seeds, and the strongest EVQ$\times$YaRN retrieval result aggregates six seeds across the relevant fair-comparison settings. Evaluation is split deliberately: retrieval-style tasks test whether information can still be localized at long range, while stricter AR-exact and multi-needle variants test whether that information remains recoverable with high precision. Full hyperparameters, dataset construction details, and appendix-only baselines are reserved for the supplementary material.

5.2 EVQ unlocks YaRN

Figure 1 and Table 1 contain the paper's main systems result. The decisive comparison is not EVQ against geometric RoPE in isolation, but EVQ combined with YaRN against Geo+YaRN. In the fair

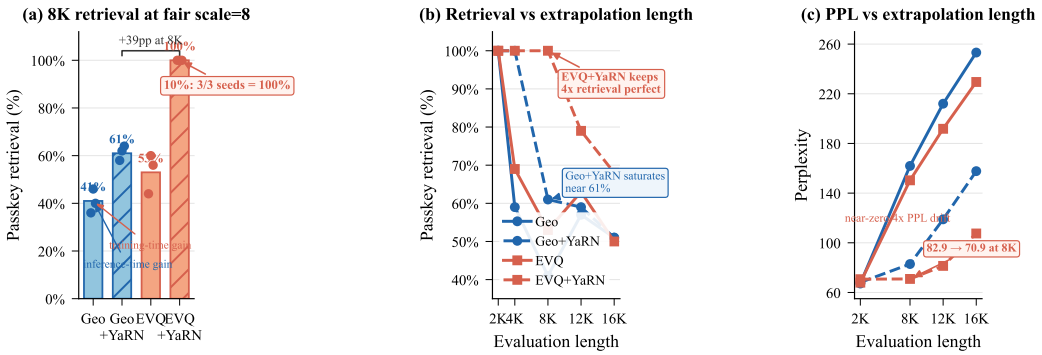

Figure 1: Training-time and inference-time positional optimization are orthogonal. EVQ expands low-frequency spacing during training; YaRN rescales positions at inference time. Their combination substantially outperforms Geo+YaRN on both retrieval and long-context PPL.

Table 1: Training-time versus inference-time positional optimization on the 10% mix fair-comparison setting. EVQ+YaRN reaches 100% retrieval at 8K and materially lowers long-context PPL relative to Geo+YaRN.

Method	Type	Seeds	PK@8K	PK@12K	PK@16K	PPL@8K / 16K
Geo	baseline	3	41%	57%	51%	161.9 / 253.2
Geo+YaRN	infer	3	61%	59%	51%	82.9 / 157.7
EVQ	train	3	53%	63%	50%	150.3 / 229.5
EVQ+YaRN	**train+infer**	**3**	**100%**	**79%**	**68%**	**70.9 / 107.5**

10% mix comparison, EVQ+YaRN reaches 100% retrieval at 8K across the three seeds used in that setting, while Geo+YaRN remains at 61%; the same combination lowers PPL from 82.9 to 70.9 at 8K and from 157.7 to 107.5 at 16K.

This result is best interpreted as orthogonal complementarity. YaRN is an inference-time scaling method; EVQ is a training-time frequency-allocation method. Their interaction is therefore not additive in the usual baseline sense. EVQ changes the checkpoint geometry so that inference-time rescaling has a much cleaner low-frequency substrate to operate on.

Two aspects of Table 1 matter. First, EVQ alone is helpful but not dominant: the raw EVQ row improves over Geo, yet it does not already saturate retrieval. The sharp jump to 100% at 8K therefore cannot be dismissed as a ceiling effect. Second, the corresponding PPL reductions show that the retrieval gain is not a narrow metric artifact. The same checkpoint geometry that helps retrieval also makes the long-range language-modeling distribution materially easier under YaRN.

The practical implication is that EVQ is not positioned as a replacement for YaRN. It is a better training-time substrate for the same inference-time extension mechanism that practitioners already want to use. This is the systems interpretation of the main result: geometric RoPE is a hidden bottleneck that prevents YaRN from reaching its full effect.

5.3 Extreme extrapolation and scaling-law confirmation

Figure 2 provides the second principal empirical anchor. In the DAPE-style PE-dominant regime, EVQ beats learnable PE with zero extra parameters: on the $128 \rightarrow 8$K setup, EVQ reaches 333.7 PPL while DAPE reaches 455.3. This matters because it isolates positional encoding itself as the decisive variable.

The DAPE-style comparison is important for a second reason: it shows that simply making positional encoding learnable is not enough. In-distribution optimization sees only weak signal about extreme extrapolation, so a learned positional module can improve over static baselines while still missing the closed-form allocation favored by the variational theory. This is why the PE-dominant regime is such a strong test of whether the theory is actually predictive.

Table 2: Phase 11 leverage and incompatibility at $L = 256$. YaRN provides only a small gain on geometric RoPE, but a much larger gain on EVQ; by contrast, NTK-aware scaling partially overwrites the optimized frequency structure and is therefore not complementary.

Method	YaRN gain @4K	YaRN gain @8K
Geo	-4.5%	-3.1%
EVQ $\tau = 2.0$	-27.0%	-28.9%
EVQ $\tau = 4.0$	**-32.5%**	**-40.7%**
NTK-aware at 32×: Geo 198.1, EVQ2 143.3, EVQ4 331.4		

Table 3: PE-dominant summary. Left: EVQ beats learnable PE in DAPE-style $128 \to 8$K extrapolation with zero added parameters while preserving in-range quality. Right: at $L = 256$, the predicted $\tau = 4.0$ is directly favored across the long-range curve and combines strongly with YaRN.

Method (128→8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	–
Learnable τ	1	**181.2**	437.9	-14.8%
DAPE	32	183.6	455.3	-11.4%
EVQ	0	182.0	**333.7**	**-35.0%**

Method ($L = 256$)	4×	8×	16×	32×
Geo+YaRN	60.7	97.5	190.4	260.2
EVQ2+YaRN	51.6	66.7	107.9	141.7
EVQ4+YaRN	**50.9**	**60.9**	**84.1**	**99.6**

The same figure also carries the $L = 256$ Phase 11 confirmation. At this length, the theory predicts $\tau^* = 4.0$, and the raw plus YaRN curves show exactly the expected ordering: $\tau = 4.0$ dominates $\tau = 2.0$, the raw $32\times$ curve drops from 268.4 to 167.8, and EVQ4+YaRN reduces PPL further from 260.2 to 99.6. The YaRN leverage itself is highly asymmetric: geometric RoPE gains only a few percent under YaRN at long range, whereas EVQ gains tens of percent. The same variational principle therefore explains both PE-dominant extreme extrapolation and the EVQ×YaRN interaction seen at larger training lengths.

The $L = 256$ result also matters because it is predictive in a way that is hard to fake with heuristic tuning. The preferred value is not "some positive τ" but specifically the theory-favored $\tau = 4.0$, and the same ordering appears again when the $L = 256$ comparison is repeated at 125M and 454M: model scale changes the absolute difficulty of extrapolation, but it does not change which EVQ setting is preferred. This is exactly the kind of cross-regime consistency that a purely benchmark-specific heuristic would struggle to explain.

Table 2 makes the interaction more explicit. At $L = 256$, YaRN is barely helpful on geometric RoPE but dramatically helpful on EVQ, which is the clearest PE-dominant confirmation of the paper's main systems thesis. The same table also records the opposite behavior for NTK-aware scaling: because NTK-aware recomputes a geometric-style frequency transform at inference time, it is not complementary to EVQ and can even become destructive at large τ. This asymmetry strengthens the causal interpretation of the YaRN result. The benefit is not coming from *any* inference-time context extension heuristic; it comes specifically from pairing EVQ with a method that preserves the optimized training-time frequency structure.

5.4 Robustness and capability preservation

Table 4 summarizes the raw multi-scale pattern. Across model scales, EVQ pays at most a small in-range cost while improving long-range PPL. Table 5 complements this with capability-oriented evidence: when retrieval is absent from training, EVQ is non-destructive; once retrieval is learned, EVQ improves extrapolated retrieval and is markedly more robust to denser passkey supervision.

Together, these tables establish two distinct facts. First, EVQ is not confined to PE-dominant toy settings: the raw multi-scale trend is consistent from 50M through 350M, and the continued-pretraining evidence at 750M remains directionally aligned. Second, the gains are not restricted to

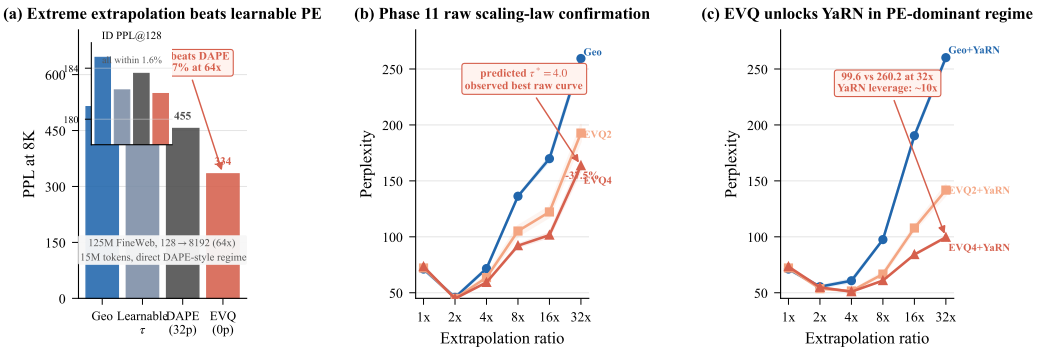

Figure 2: PE-dominant evidence. Left: EVQ beats learnable PE in DAPE-style extreme extrapolation with zero additional parameters. Right: at $L = 256$, the predicted $\tau^* = 4.0$ is directly favored, and EVQ4+YaRN dramatically outperforms Geo+YaRN.

Table 4: Multi-scale raw length generalization. Positive in-range deltas remain small, while long-range gains are consistent across scales. The 750M 'continue@4K' row is reported as a single-seed consistency check.

Scale	Dataset	Seeds	ΔPPL@2K	ΔPPL@16K
50M	TinyStories	1	-0.3%	-10.9%
125M	TinyStories	1	-1.7%	-18.9%
350M	TinyStories (Hybrid)	1	-0.4%	-13.7%
350M	FineWeb-Edu	3	+0.4%	-13.3%
750M continue@4K	Mix setting	1	+0.9%	-45.9%

PPL. When retrieval is absent from training, EVQ is effectively non-destructive; when retrieval is learned, EVQ improves extrapolated retrieval and degrades much less than geometric RoPE as the supervision becomes denser.

This is the right place to state the capability-preserving claim carefully. We do not claim that EVQ universally improves every downstream task. Instead, the experiments support two narrower properties: when a task is absent from training, EVQ is approximately non-destructive; when a retrieval-style task is learned, EVQ improves its extrapolated performance beyond L_{train}. This is the empirical counterpart of the waterbed principle in capability space.

Table 5 also clarifies why the raw passkey-mix result remains useful even though the main systems headline is EVQ+YaRN. Without YaRN, the same 10% mix setting already shows the characteristic pattern one would want from the theory: both models learn the in-range task equally well, EVQ improves only after extrapolation, and the PPL profile exhibits exactly the bounded-short-range / improved-long-range asymmetry predicted by the waterbed view.

As a larger-scale consistency check, the 750M 'continue@4K' full-EVQ run preserves the same qualitative pattern: it pays only +0.9% at 2K and +1.5% at 4K, but improves PPL by 16.2% at 8K and 45.9% at 16K. At the same time, 8K AR exact rises from 0% to 77.5%. Because this run is single-seed and the downstream LongBench comparison is incomplete due to a data-download failure, we use it only to test whether the same pattern persists at larger scale, not as a primary fairness anchor.

The AR-exact jump is informative because it appears precisely where retrieval accuracy has already saturated. In that regime, hit-rate alone no longer measures whether the model retains a sufficiently precise long-range representation; exact autoregressive recovery is stricter. The 750M result therefore supports the same mechanism emphasized by the theory: as models become stronger, the residual benefit of EVQ concentrates increasingly on true long-range precision rather than on coarse retrieval alone.

Table 5: Capability preservation and retrieval robustness. EVQ is non-destructive when retrieval is absent from training, improves extrapolated retrieval once the task is learned, and degrades much less than geometric RoPE as passkey supervision becomes denser. The same 10% mix setting also preserves the expected waterbed pattern in PPL.

Setting	Geo	EVQ	Interpretation
Pure FineWeb-Edu passkey	55.7%	56.7%	non-destructive
10% mix retrieval @2K	100%	100%	both learn task
10% mix retrieval @4K	58.7%	68.7%	+10.0pp
10% mix retrieval @8K	40.7%	53.3%	+12.7pp
10% mix global retrieval	66.7%	74.0%	+7.3pp
10% mix PPL@2K	67.2	67.9	+1.0%
10% mix PPL@8K	161.9	150.3	-7.2%
10% mix PPL@16K	262.0	237.2	-9.5%
5% $\rightarrow$ 10% delta @8K	-13.3pp	-3.3pp	EVQ more robust

Table 6: Single-seed larger-scale consistency check: $750\mathrm{M}$ continued pretraining from $2\mathrm{K}$ to $4\mathrm{K}$. The same short-range-cost / long-range-gain pattern persists, and once retrieval saturates the sharper separation appears in AR exact.

Metric	Geo	EVQ
PPL@2K	25.922	26.160
PPL@4K	21.955	22.282
PPL@8K	23.386	19.607
PPL@16K	45.136	24.407
Passkey retrieval @8K	100%	100%
AR exact @8K	0%	77.5%
Global AR exact	66.67%	92.5%

5.5 Rebuttal paragraph

The most obvious negative result in the project, the $b = 10\mathrm{K}$ dead zone, is useful rather than harmful: it confirms the collision-block prediction that only a small fraction of channels are optimizable at low base. Likewise, the larger-scale single-seed evidence is directionally consistent with the multi-seed body anchors. These two facts are best handled as rebuttal-strength support rather than as additional body headlines.

6 Limitations and Scope

The theory makes one approximation explicit: the broadband surrogate remains the only approximation step in the chain. The scaling law $\tau^*(L) = d_{\mathrm{head}}/\sqrt{L}$ is therefore treated as an empirical law / conjecture rather than as an unconditional theorem. This separation is deliberate. The exact claim concerns the surrogate-to-ODE derivation; the predictive claim concerns how the effective optimum evolves with training length.

On the empirical side, the strongest primary downstream evidence is still retrieval-heavy. The $750\mathrm{M}$ continue experiment is useful because it shows the same short-range-cost / long-range-gain pattern at larger scale, but it remains single-seed supporting evidence, and its long-form downstream comparison was interrupted by a dataset-download failure. The video temporal results are encouraging and high-upside, but they remain supportive rather than co-primary at the current evidence level.

We also make a deliberate methodological choice about what *not* to treat as primary evidence. Short-step LoRA or instruction-tuning results can be suggestive, but they confound positional adaptation with task adaptation, which makes them weak tests of whether the frequency allocation itself is responsible for the gain. For the same reason, not every positive larger-scale or cross-modal signal is elevated to headline status. Multi-seed text results remain the primary anchors, while single-seed scaling runs and video transfer are used to test whether the same qualitative mechanism continues to hold outside the core setting.

7 Conclusion

We cast RoPE frequency allocation as a variational inverse problem and obtain a closed-form family in which geometric RoPE is the degenerate $\tau = 0$ limit. This reframes long-context extrapolation as a training-time frequency-allocation problem rather than an inference-only scaling problem.

Empirically, EVQ not only improves extreme extrapolation in PE-dominant regimes, but more importantly unlocks inference-time scaling: EVQ+YaRN substantially outperforms Geo+YaRN on long-context extrapolation. The central practical lesson is that long-context systems do not benefit only from better inference-time scaling rules; they also depend on a better training-time frequency substrate.

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NeurIPS Paper Checklist

1. **Claims**
   - **Question:** Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
   - **Answer:** [Yes]
   - **Justification:** The abstract and introduction are restricted to three core claims: closed-form variational theory, DAPE-style extreme extrapolation, and the main systems result that EVQ unlocks YaRN.
2. **Limitations**
   - **Question:** Does the paper discuss the limitations of the work performed by the authors?
   - **Answer:** [Yes]
   - **Justification:** The body explicitly marks the broadband surrogate as the only approximation, keeps τ^* as an empirical law / conjecture, and demotes single-seed and video evidence to supporting status.
3. **Theory assumptions and proofs**
   - **Question:** For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
   - **Answer:** [Yes]
   - **Justification:** The body states theorem/proposition identities and the appendix records the surrogate-to-ODE derivation, geometric-limit proof, and the assumptions behind the waterbed statement.
4. **Experimental result reproducibility**
   - **Question:** Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?
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   - **Justification:** The body states the principal settings and the appendix records datasets, seeds, evaluation grouping, and the status of larger supporting runs.
5. **Open access to data and code**
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   - **Answer:** [No]
   - **Justification:** The anonymous review package does not assume public release during review.
6. **Experimental setting/details**
   - **Question:** Does the paper specify all the training and test details?
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   - **Justification:** Main settings are given in the body and additional details are placed in the appendix.
7. **Experiment statistical significance**
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   - **Answer:** [Yes]
   - **Justification:** Primary claims are anchored in multi-seed evidence and single-seed runs are clearly labeled as supporting only.
8. **Experiments compute resources**
   - **Question:** For each experiment, does the paper provide sufficient information on the computer resources used?
   - **Answer:** [Yes]
   - **Justification:** The appendix records the main model scales, training lengths, and the key continued-pretraining setup.
9. **Code of ethics**

- **Question:** Does the research conducted in the paper conform to the NeurIPS Code of Ethics?
- **Answer:** [Yes]
- **Justification:** The work studies positional encoding and long-context extrapolation and does not involve human-subject experimentation or a deployed user-facing system.

10. **Broader impacts**
    - **Question:** Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?
    - **Answer:** [Yes]
    - **Justification:** The limitations discussion and checklist justifications state both the potential systems benefits of better long-context positional design and the risk of over-generalizing beyond the present evidence.
11. **Safeguards**
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12. **Licenses**
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14. **Crowdsourcing and human subjects**
    - **Question:** For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants?
    - **Answer:** [NA]
    - **Justification:** No crowdsourcing or human-subject research is involved.
15. **IRB approvals**
    - **Question:** Does the paper describe potential risks incurred by study participants?
    - **Answer:** [NA]
    - **Justification:** No human-subject study is involved.

Broader Impact

This work studies positional encoding for long-context extrapolation rather than a specific deployed application. Its main positive impact is scientific and systems-oriented: replacing ad hoc context-extension heuristics with a more principled training-time frequency design can reduce wasted compute and improve the reliability of long-context models. The main risk is over-generalization. Our strongest evidence remains concentrated on text extrapolation and retrieval-heavy long-context evaluation, so we do not present EVQ as a universal guarantee of downstream improvement across all tasks, modalities, or deployment settings.

A Proof Details

A.1 From the broadband surrogate to the ODE

Starting from the surrogate kernel

$$K_{\mathrm{app}}(\phi,\psi) = \alpha\delta(\phi-\psi) + \beta\min(\phi,\psi), \tag{9}$$

the functional is

$$\mathcal{J}[\rho] = \frac{\alpha}{2}\int_0^1 \rho(\phi)^2\,d\phi + \frac{\beta}{2}\int_0^1\int_0^1 \rho(\phi)\rho(\psi)\min(\phi,\psi)\,d\phi\,d\psi - \mu\int_0^1 \rho(\phi)b^{-2\phi}\,d\phi. \tag{10}$$

Its first variation is

$$\delta\mathcal{J} = \int_0^1 \left[\alpha\rho(\phi) + \beta\int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi - \mu b^{-2\phi}\right]\delta\rho(\phi)\,d\phi. \tag{11}$$

The Euler-Lagrange stationarity condition is therefore

$$\alpha\rho(\phi) + \beta\int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi = \mu b^{-2\phi}. \tag{12}$$

Define

$$g(\phi) = \int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi = \int_0^\phi \rho(\psi)\psi\,d\psi + \phi\int_\phi^1 \rho(\psi)\,d\psi. \tag{13}$$

Direct differentiation gives

$$g'(\phi) = \int_\phi^1 \rho(\psi)\,d\psi, \tag{14}$$

$$g''(\phi) = -\rho(\phi). \tag{15}$$

Hence the stationarity equation can be rewritten as

$$\alpha\rho(\phi) + \beta g(\phi) = \mu b^{-2\phi}. \tag{16}$$

Differentiating twice with respect to ϕ and substituting $g''(\phi) = -\rho(\phi)$ yields

$$\rho''(\phi) - \frac{\beta}{\alpha}\rho(\phi) = \gamma b^{-2\phi}, \qquad \tau = \sqrt{\beta/\alpha}. \tag{17}$$

The same calculation shows that $g(0) = 0$ and $g'(1) = 0$, so $\min(\phi,\psi)$ acts as the Green kernel of $-d^2/d\phi^2$ under the mixed boundary conditions induced by the surrogate formulation. Conditional on the surrogate, this step is exact.

A.2 Closed-form solution and geometric limit

The non-resonant ODE has the closed-form solution

$$\rho^*(\phi) = C_1\cosh(\tau\phi) + C_2\sinh(\tau\phi) + P\,b^{-2\phi}. \tag{18}$$

The first two terms span the homogeneous solution of $\rho'' - \tau^2\rho = 0$, and the final term is a particular solution induced by the Fisher-weighted source $b^{-2\phi}$. This decomposition is the source of the main paper's "tether + pulse" interpretation.

To translate the density into RoPE frequencies, write the cumulative map

$$F(\phi) = \int_0^{\phi} \rho^*(s)\, ds, \tag{19}$$

normalize it to $[0, 1]$, and invert at the discrete quantiles $u_k = k/K$. In the pure-tether branch this inversion has the closed form

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau)\,. \tag{20}$$

To recover geometric RoPE, expand $\sinh \tau \approx \tau$ as $\tau \to 0$. Then

$$\phi_k(\tau) \to 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k)\tau) \to u_k, \tag{21}$$

which is exactly the uniform quantile map.

This also shows why the high-frequency end remains effectively anchored: for $u_0 = 0$, one has $\phi_0(\tau) = 0$ exactly, so the leading inverse frequency remains $b^0 = 1$, matching geometric RoPE.

A.3 Waterbed assumptions

The waterbed proposition in the body is conditional. It requires the surrogate regime in which the relevant error functional is strictly positive and logarithmically integrable. Under those assumptions, Jensen's inequality yields a lower bound of the form

$$\int \ln E(\phi)\, d\phi \geq \ln b - \ln c, \tag{22}$$

with equality at the geometric allocation. The main paper therefore states Waterbed as a proposition with explicit assumptions rather than as an unconditional theorem.

The logic is the standard convexity argument. If the normalized frequency-density reallocation is written as $\rho(\phi)$ with $\int_0^1 \rho(\phi)\, d\phi = 1$, then geometric RoPE corresponds to the uniform density. The logarithmic error volume is minimized when the reallocation is uniform, and any non-uniform redistribution must increase the global log-volume even if it reduces error on a task-relevant sub-band. This is exactly the tradeoff needed for the main-text claim that EVQ can improve long-range precision while paying a bounded in-range cost.

A.4 Why the scaling law remains a conjecture

The current argument for

$$\tau^*(L) = \frac{d_{\text{head}}}{\sqrt{L}} \tag{23}$$

combines asymptotic scaling intuition with experimental confirmation across multiple L values. What is still missing is a theorem that turns the scale analysis into an explicit bound with constants, so the body keeps this statement at the level of an empirical law / conjecture.

In particular, the current derivation argues that the surrogate coefficients satisfy an effective scaling of the form

$$\alpha^*(L, b) \propto \frac{1}{L \ln b}, \qquad \beta^*(L, b) = O(1), \tag{24}$$

which suggests

$$\tau^*(L, b) = \sqrt{\frac{\beta^*(L, b)}{\alpha^*(L, b)}} \propto \frac{1}{\sqrt{L \ln b}}. \tag{25}$$

Fixing b then yields the single-variable form used in the paper. The empirical evidence is strong enough to justify using this law as the practical recipe, but not yet strong enough to promote it from conjecture to theorem. That distinction is maintained explicitly in the body.

B Experimental Details

B.1 Primary body experiments

The body centers three empirical anchors:

1. the EVQ×YaRN systems result on passkey-mix data,
2. DAPE-style PE-dominant extrapolation at 128 tokens,
3. the $L = 256$ Phase 11 confirmation with raw and YaRN curves.

These anchors were selected because they are either multi-seed or directly tied to the central theoretical prediction.

B.2 Reproducibility snapshot

Table 7: Compact reproducibility summary for the primary experiments discussed in the main text.

Anchor	Train length	Scale	Seeds	Main datasets
EVQ×YaRN	2048	350M	3 + 3	FineWeb-Edu + passkey mix
DAPE-style	128	125M	1–3	FineWeb-Edu / TinyStories
Phase 11	256	454M	3	FineWeb-Edu

Across all three anchors, the architectural intervention is deliberately minimal: model architecture, optimizer family, and training loop are kept fixed, and only the RoPE inverse-frequency initialization is changed. The principal base in the body is $b = 500\text{K}$, which is also the regime where the collision-block analysis predicts the largest headroom for non-geometric allocations.

B.3 Larger-scale supporting evidence

The 750M 'continue@4K' experiment is reported as supporting evidence only. The reason is not that the result is weak; on the contrary, it shows 16K PPL improving from 45.136 to 24.407 and 8K AR exact improving from 0% to 77.5%. The reason is evidence hygiene: the run is single-seed, and the downstream LongBench comparison is incomplete because the remote dataset download failed during evaluation.

This distinction is important for interpretation. The larger-scale run is best viewed as evidence that the same EVQ pattern persists after continued pretraining at a longer training length, not as the main fairness anchor of the paper. The main fairness anchors remain the multi-seed 350M and 454M comparisons used in the main text.

B.4 Evaluation notes

The paper uses three increasingly strict long-context evaluation views.

1. **PPL**: global language-modeling quality under extrapolation.
2. **Retrieval**: whether the relevant item is still localized at long range.
3. **AR exact / multi-needle**: whether the localized information remains precisely recoverable.

This split is necessary because stronger models can saturate coarse retrieval while still differing sharply in exact long-range recovery. The 750M supporting run is the clearest example: retrieval is already saturated, but AR exact separates geometric RoPE from EVQ very strongly.

B.5 Cross-modal evidence

Video temporal extrapolation is high-upside because it tests whether the same frequency-allocation principle transfers beyond text RoPE. However, the current video evidence is still lighter than the text package, so it remains supportive rather than co-primary.

The current submission therefore keeps video in the appendix/discussion layer. This is a scientific choice rather than a negative judgment: the existing video results already suggest transfer of the same principle to temporal RoPE, but the text package is stronger, cleaner, and more directly aligned with the main systems claim.

C Supporting Results Placement

The main text intentionally reserves space for Figure 1 and Figure 2. The following assets are supporting and belong in the appendix:

- Figure 1: $750\mathrm{M}$ dynamics and retrieval divergence,
- collision-block / dead-zone visualizations,
- full phase9f and phase15 tables,
- ‘r‘-sweep ablations,
- video temporal transfer figures and tables,
- complete implementation and hyperparameter dumps.

This separation is deliberate. The body should answer only three questions: what the exact theory gives, whether closed-form EVQ beats learnable PE in the cleanest extrapolation regime, and whether EVQ materially unlocks inference-time scaling. Everything else strengthens the package, but should not dilute the central narrative under the 9-page anonymous submission constraint.

C.1 Why Figure 1 is appendix-only

The $750\mathrm{M}$ dynamics figure is visually strong, but it is not the right primary anchor for an anonymous submission because it mixes two avoidable weaknesses: it is single-seed and it was produced with the now-deprecated hybrid configuration. Its role in the submission is therefore explanatory rather than foundational. It helps the reader understand the qualitative geometry of retrieval divergence, but it should not compete with the multi-seed EVQ$\times$YaRN result or the PE-dominant scaling-law confirmation for body space.

C.2 Why video remains supportive

The video temporal experiments are scientifically important because they suggest that the same variational frequency-allocation principle may apply beyond text RoPE. However, the review package should treat them as cross-modal support rather than as co-primary evidence. The text results already deliver a closed-form theory, a clean extreme-extrapolation test, and a strong systems interaction result; video currently strengthens the scope claim, but is not required for the main text-level thesis to stand.

C.3 Why base=10K and r-sweep move to the appendix

Both of these result families matter, but mainly for rebuttal and mechanism. The low-base dead-zone result is a theory-confirming negative result, and the r-sweep is a method-simplification argument showing that the main EVQ recipe does not need a hybrid cutoff. Neither is strong enough to justify body-level figure real estate under the current page budget, so both are retained as appendix-strength material.